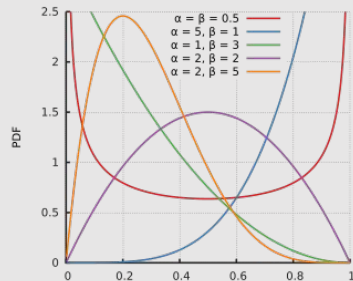

Welcome to CS 109ACE!

Ethan Hersch

Hi, I'm Ethan!



Deep learning, AI, statistics



Agenda

1. Logistics + intros
2. Lecture review - Counting
 - a. Practice!
3. Lecture review - Intro Probability
 - a. More practice!
4. Lecture review - Bayes Theorem
 - a. Practice (again)!
5. + your questions

Let's get to know each other!

Name, year, major

Where are you from?

1 thing you miss the most about the summer

1 thing you're most excited about this quarter

First some logistics...

We will meet every Wednesday

Attend 7/9 sessions to receive credit for the quarter

- Join the slack! <https://cs109ace.slack.com/archives/C0C4PGF59FX>
 - Can reach out whenever
 - Use as a resource to ask general questions about the course
- Fill out OH availability form! <https://www.when2meet.com/?38886965-Tb3sy>
- 1:1 meetings with me at least once in the quarter (reach out if other concerns)

One other form 🤪

Intro form, other information – please complete before next week!

https://docs.google.com/forms/d/e/1FAIpQLSe1mAsUZwbRjNByQ0WeNJBttXR4dU4jgqT3zDxu_NWCYBJtMQ/viewform?usp=publish-editor

(I'll send in the Slack, along with all the other forms)

ACE Mission

“The mission of ACE is to provide our students, particularly those from under-resourced and/or minoritized backgrounds, with the opportunities and support to succeed in their course and in their major. ACE sections, instruction, and the ACE Course Assistants (CA’s) are intentional about fostering environments based on equity, purposeful engagement, and inclusive teaching practices.”

Expectations from ACE

Attend lectures for CS 109

(Remember that ACE is a supplementary course and not a replacement for the course lectures)

In 109, ACE does not replace regular section!

Attend and be engaged in section

Make the most out of their time in ACE!

Expectations

Ask questions! I'm not here to grade you, lecture you, etc. This course is meant for support.

Tell me if something isn't working: this class will be tuned to your preferences throughout this quarter (surveys, exit tickets, check-ins)

Make mistakes. I'll make mistakes this quarter, and we all will.

My Teaching Philosophy

Nobody starts an expert, and nothing replaces hard work.

This all comes from making mistakes and not being afraid to try new problems.

We will try a lot of new problems together and encourage mistakes!

Why CS 109?

First, we will do combinatorics:

A surprising birthday question 🎂

How many people need to be in a room before there's a greater than 50% chance that two share a birthday?

Guesses?

Solution

A general case: for n people

$$P(\text{no share a birthday}) = (365/365) (364/365) \dots (365-n+1)/365$$

If $n = 22$, this probability is 52.4%

If $n = 23$, this probability is 49.3%

Why CS 109?

Then we will do more traditional probability.

You're collecting all 10 different stickers. Each pack contains one random sticker. On average, how many packs will it take to get all 10?

Solution

$$E[T] = 10/10 + 10/9 + \dots + 10/1 = \sim 29/29$$

How did we get here?

If we have k different stickers, the chance a pack has a new one is $(10-k)/10$

The average number of packs needed to go from k to $k+1$ distinct stickers is $10/(10-k)$

Why CS 109?

Some statistics as well.

How many coin flips until the average looks fair?

A fair coin is flipped 100 times. Let X be the number of heads.

What's the approximate probability of getting between 40 and 60 heads?

Solution

The coin is flipped 100 times. $X \sim \text{Binomial}(100, 0.5, 0.5)$

What's its EV? $E[X] = 50$

What's its standard deviation? $= \sqrt{npq} = \sqrt{100 * 0.5 * 0.5} = 5$

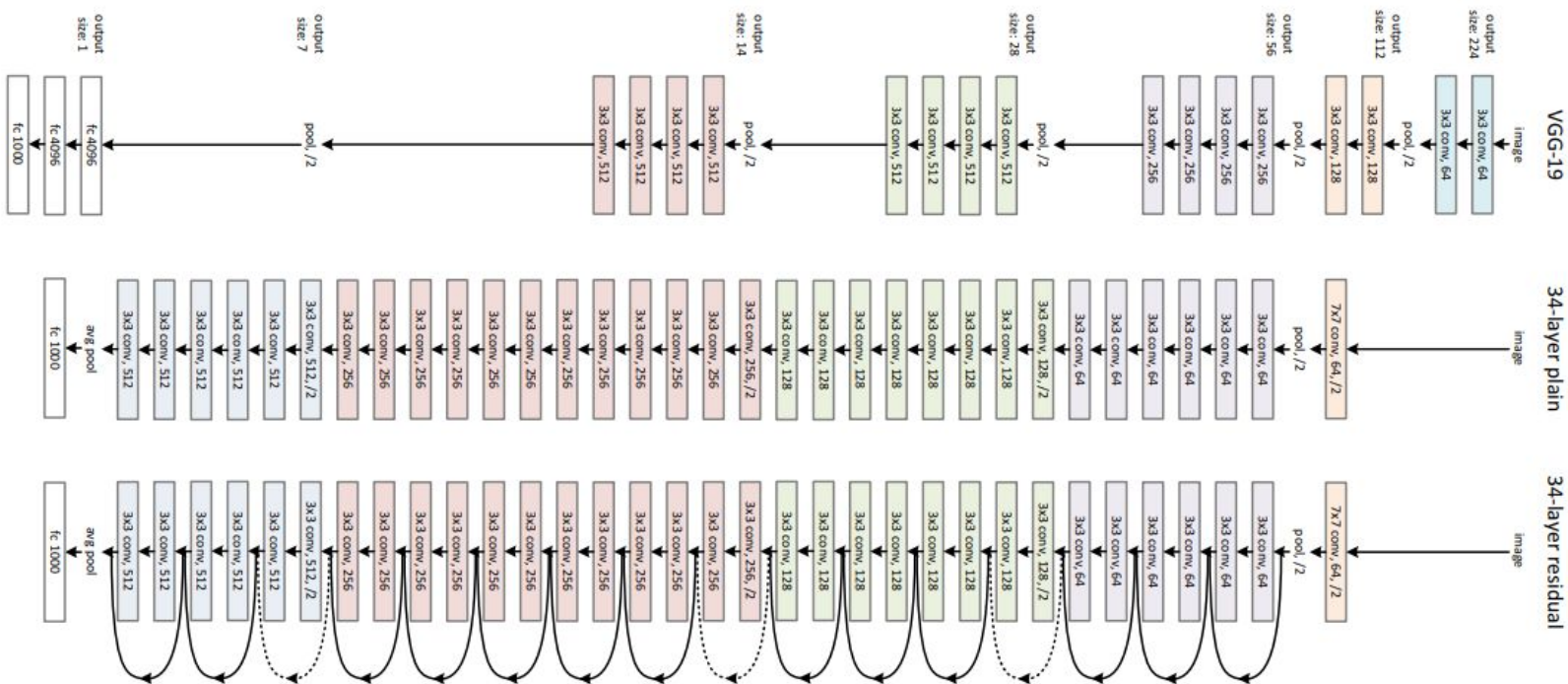
$P(40 \leq X \leq 60)$? $P(-2 \leq Z \leq 2)$ within 2 stdevs **95.45%**

This uses the Central Limit Theorem

Why CS 109?

Then finally, we will cover machine learning!





One last problem... work with me!

Let X_i be the time at which you see the i -th new word.

Example: Pushing six times yields: penguin, penguin, randomized, hello, randomized, spaceship
Then $X_1 = 1$, $X_2 = 3$, $X_3 = 4$, and $X_4 = 6$.

Group Work

1. What is $E[X_1]$? *(This is not a trick question)*
2. What is $E[X_2 - X_1]$? *(In expectation, how many pushes after the 1st word before seeing a 2nd new word?)*
3. What is $E[X_3 - X_2]$?
4. For any $i = 2, 3, \dots, n$, what is $E[X_i - X_{i-1}]$?
5. Use your answers above + linearity of expectation: what is the expected number of pushes to see all n words? *(Try to simplify to a expression)*

What do you want to get out of this?

If you've done ACE before, what worked? What would you like to change?

How much independent work / practice would you like compared to review?

Other things you want?

Python review?

Homework tips?

General Course Tips

Structure

Everything builds off of earlier concepts. Make sure you understand these things really well

Chris is great, so don't be afraid to ask questions, go to OH, etc.

PSET Collaboration

I'm not here to give HW answers

If there are concepts from the HW you want to discuss, let's bring them up!

You can discuss concepts with this group (and form your own groups)!

DO NOT share/look at written solutions (talk through at high level)

Disclose who you worked with

AI Policy

AI will do this class for you.

The PSETs barely count and are the best practice for the test, so remember it is not worth it

These concepts are actually important and fun!

But...

You can use AI as a great study tool!

- Make practice problems
- If you have an idea for a fun project, try it out!
- Use it to answer your questions
- Talk with your favorite assistant before figuring out what questions to ask!
- If I prompt you for a question during class, I might ask you to ask an LLM to give a good question

Practice Asking Questions

That's an objective of this course: be comfortable with asking questions

It's such an important skill

Onto the material...

Topics Covered Last Week:

- Intro to probability
- Conditional probability

- Bayes' Theorem
- Counting and combinatorics

Specifics to add?

Any specific topics or portions you want to focus on?

Intro to probability

What Is Probability?

Axioms of Probability

Recall: S = all possible outcomes, E = the event.

- Axiom 1: $0 \leq P(E) \leq 1$
- Axiom 2: $P(S) = 1$
- Axiom 3: If events E and F are mutually exclusive,

$$P(E \cup F) = P(E) + P(F)$$



Andrey Kolmogorov

Sample Spaces and Events

Sample space S : the set of all possible outcomes of an experiment

Event E : any subset of S ($E \subseteq S$)

One coin: $S = \{H, T\}$. Two coins: $S = \{HH, HT, TH, TT\}$

One die: $S = \{1, \dots, 6\}$; “roll is even” is $E = \{2, 4, 6\}$

More Notation

$E \cup F = E \text{ or } F$. $E \cap F = EF = E \text{ and } F$. $E^c = \text{not } E$

De Morgan: $(E \cup F)^c = E^c \cap F^c$ and $(E \cap F)^c = E^c \cup F^c$

Consequences of the Axioms (draw out visually)

Complement: $P(E^c) = 1 - P(E)$ (E and E^c are disjoint and make up S)

If E and F are mutually exclusive, $P(E \cup F) = P(E) + P(F)$

Inclusion–exclusion: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

If $E \subseteq F$, then $P(E) \leq P(F)$. $P(\emptyset) = 0$

Key idea: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ $P(E^c) = 1 - P(E)$

Quick Practice

What is the probability of rolling at least 1 6 in 4 rolls of a fair dice?

Solution

$$P(\text{at least one 6 in 4 rolls}) = 1 - (5/6)^4 \approx 0.518$$

1. Find the probability of not rolling a 6 on a single roll
2. Find the probability of not rolling a 6 in all 4 independent rolls
3. Subtract from 1 to find the probability of at least one 6

Why is this important?

Example: roll a fair 6-sided die.

$S = \{1,2,3,4,5,6\}$. Let $E = \{\text{roll a 1}\}$ and $F = \{\text{roll a 6}\}$.

1. Probability is between 0 and 1: $P(E) = \frac{1}{6}$
2. Something in the sample space must happen: $P(S) = 6/6 = 1$
3. Add mutually exclusive events: A roll can't be both 1 and 6, so

$$P(E \cup F) = P(E) + P(F) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Equally Likely Outcomes

If every outcome in S is equally likely: $P(E) = |E| / |S|$

So probability turns into counting: count the event, count the sample space

Die: $P(\text{even}) = |\{2, 4, 6\}| / 6 = 1/2$

Card: $P(\text{heart}) = 13/52 = 1/4$

Count E and S the same way (both ordered or both unordered)

Only valid when outcomes really are equally likely (sums of two dice aren't!)

Key idea: $P(E) = |E| / |S|$, but only when every outcome is equally likely/uniform random

Let's do it together

Roll two dice. What outcomes make up 'sum is 7'? What is its probability?

Sum Is 7: Answer

Treat the dice as distinct (die 1, die 2): $|S| = 6 \cdot 6 = 36$ equally likely outcomes

$E = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$, so $|E| = 6$

$$P(E) = |E| / |S| = 6/36 = 1/6$$

Trap: the 11 sums (2 to 12) are NOT equally likely

$$P(\text{sum} = 2) = 1/36, \text{ but } P(\text{sum} = 7) = 6/36$$

Answer: $P(\text{sum} = 7) = 6/36 = 1/6$

Counting Basics: Sum & Product Rules

Product rule: m outcomes and n outcomes for each one $\rightarrow m \cdot n$ in total

6-digit entry codes: $10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 1,000,000$

Sum rule: if A and B don't overlap, $|A \cup B| = |A| + |B|$

Inclusion–exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$

6 paths through Kilimanjaro, 8 through Mombasa, 2 through both

$\rightarrow 6 + 8 - 2 = 12$ paths

Counting Basics: Order Matters?

Does order matter? Are the objects distinct?

Arrange n distinct objects in a row: $n!$

Ordered picks of k from n (permutations): $n! / (n - k)!$

Unordered picks of k from n (combinations): $C(n, k) = n! / (k! (n - k)!)$

3 of 4 distinct marbles: ordered $4 \cdot 3 \cdot 2 = 24$; unordered $24 / 3! = 4$

Identical objects: divide out the repeats

BASKETBALL: $10! / (2! 2! 2!) = 453,600$

Key idea: order matters $\rightarrow n! / (n - k)!$ order doesn't $\rightarrow C(n, k) = n! / (k! (n - k)!)$

Conditional Probability

Conditional Probability

Def. The conditional probability of E given F is the probability E occurs given that F already occurred. Sometimes this is also “conditioning on F .”

$$P(E \mid F) =$$

$$= (\# \text{ outcomes in } E \text{ consistent with } F) / (\# \text{ outcomes in } S \text{ consistent with } F)$$

$$= |E \cap F| / |S \cap F|$$

$$= |E \cap F| / |F|$$

Conditional Probability: Intuition

Conditioning on F shrinks the sample space down to F

$$P(E \mid F) = P(E \cap F) / P(F) \quad (\text{only defined when } P(F) > 0)$$

Two dice: E = sum is 4, F = first die is 2

$P(E) = 3/36 = 1/12$, but $P(E \mid F) = 1/6$: only (2,2) of F 's 6 outcomes

8-sided die: A = even, B = roll $\geq 6 = \{6, 7, 8\}$

$$P(A \mid B) = |\{6, 8\}| / |\{6, 7, 8\}| = 2/3$$

Key idea: $P(E \mid F) = P(EF) / P(F)$

Inside the World of F: Same Rules

After conditioning on F, every probability rule still holds:

$$P(E^c \mid F) = 1 - P(E \mid F)$$

$$P(E \cup G \mid F) = P(E \mid F) + P(G \mid F) - P(E \cap G \mid F)$$

Can we draw these?

The Chain Rule

Super important for language modeling (aside)!

$$P(E F) = P(E | F) \cdot P(F)$$

This goes the other way too!

$$P(E F) = P(F | E) \cdot P(E)$$

And with more events

$$P(E F G) = P(E) \cdot P(F | E) \cdot P(G | E, F)$$

Chain Rule: Examples

Hungry puppy (lecture): $P(H) = 0.5$, $P(K \mid H) = 0.5$

$$P(H \text{ K}) = P(H) \cdot P(K \mid H) = 0.5 \cdot 0.5 = 0.25$$

Three aces in a row from a shuffled deck (no replacement):

$$\begin{aligned} P(A_1 A_2 A_3) &= P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 A_2) \\ &= 4/52 \cdot 3/51 \cdot 2/50 = 1/5525 \end{aligned}$$

Key idea: $P(E_1 E_2 E_3) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 E_2)$, and so on for more events

How can we use the chain rule for LLMs?

What is the modeling task given some text corpus $D = \text{"The dog ran over the fence then jumped into the lake for a swim?"}$

You want to maximize the likelihood of D :

$$P(D) = P(\text{"The"}) * P(\text{"dog"} | \text{"The"}) * P(\text{"ran"} | \text{"The", "dog"}) * \dots$$

After conditional probability, what changed when we learned something

Question 1

“Given that the sum of two dice is 7, what is the probability the first die is 2?”

Question 2

“Given that the first die is 2, what is the probability the sum is 7?”

After conditional probability, what changed when we learned something

Question 1

“Given that the sum of two dice is 7, what is the probability the first die is 2?”

Result: **$1/6$**

Question 2

“Given that the first die is 2, what is the probability the sum is 7?”

Result: **$1/6$**

Did those answers have to match?

This tees up Bayes' Theorem

Bayes' Theorem

Bayes' Theorem (my favorite, and everyone's)

The #1 rule to remember from this class... can we derive this?

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

Posterior

$P(A|B)$: Updated belief after evidence

Likelihood & Prior

$P(B|A) \cdot P(A)$: Evidence fit & initial belief

Marginal

$P(B)$: Total probability of evidence

the theory that would not die

the theory



that would



not die



how bayes' rule cracked



the enigma code,

hunted down russian

submarines & emerged

triumphant from two



centuries of controversy

sharon bertsch mcgrayne

Law of Total Probability

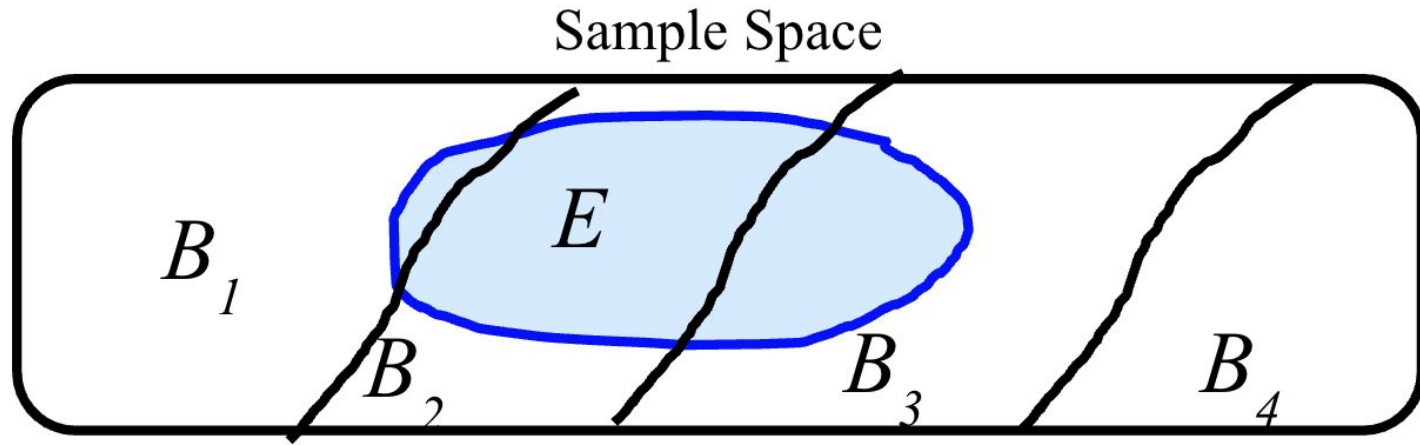
Law of Total Probability (often need with Bayes' rule)

Split E into pieces using an event M you can condition on:

$$\begin{aligned} P(E) &= P(EM) + P(EM^c) && (M^c = \text{"not M"}) \\ &= P(E \mid M) \cdot P(M) + P(E \mid M^c) \cdot P(M^c) \end{aligned}$$

General: if $B_1, B_2, \dots$ partition S, $P(E) = \sum_i P(E \mid B_i) \cdot P(B_i)$

A Good Visual



Law of Total Probability Example

Ex (lecture): 10% of bacteria have a mutation (M). Survival (E): 20% if M, 1% if not.

$$P(E) = (0.20)(0.10) + (0.01)(0.90) = 0.020 + 0.009 = 0.029$$

Followup: can we draw this visually?

LOTP: Think in Trees

Branch first on what you can condition on (M or M^c), then on E

Multiply along a branch (chain rule): $P(E | M) = P(M) \cdot P(E | M)$

Add up the branches that end in E (LOTP)

Bacteria: M (0.10) $\rightarrow$ survive (0.20) = 0.020

M^c (0.90) $\rightarrow$ survive (0.01) = 0.009

$P(E) = 0.029$: a weighted average of 20% and 1%

Branches must partition S : no overlaps, nothing missing

Key idea: $P(E) = \sum_i P(E | B_i) P(B_i)$ for any partition $B_1, B_2, \dots$

Deriving Bayes' Theorem

Chain rule, both ways: $P(M \cap E) = P(M | E) \cdot P(E) = P(E | M) \cdot P(M)$

Divide by $P(E)$: $P(M | E) = P(E | M) \cdot P(M) / P(E)$

The denominator usually comes from LOTP:

$$P(E) = P(E | M) \cdot P(M) + P(E | M^c) \cdot P(M^c)$$

Bacteria: $P(M | E) = (0.20)(0.10) / 0.029 \approx 0.69$

Surviving raises our belief in “mutation” from 10% to 69%

Bayes needs no independence assumption: it's just the definition, rearranged

Key idea: $P(M | E) = P(E | M) P(M) / P(E)$

Bayes Vocabulary: Spam Example

60% of email is spam (S). 20% of spam contains “Dear” (D); 1% of non-spam does.

Prior: $P(S) = 0.60$ (belief before seeing the email)

Likelihood: $P(D | S) = 0.20$ (how well “spam” explains the evidence)

Normalizing constant: $P(D) = (0.20)(0.60) + (0.01)(0.40) = 0.124$

Posterior: $P(S | D) = 0.12 / 0.124 \approx 0.97$ (belief after the evidence)

Key idea: posterior = likelihood $\times$ prior / normalizing constant

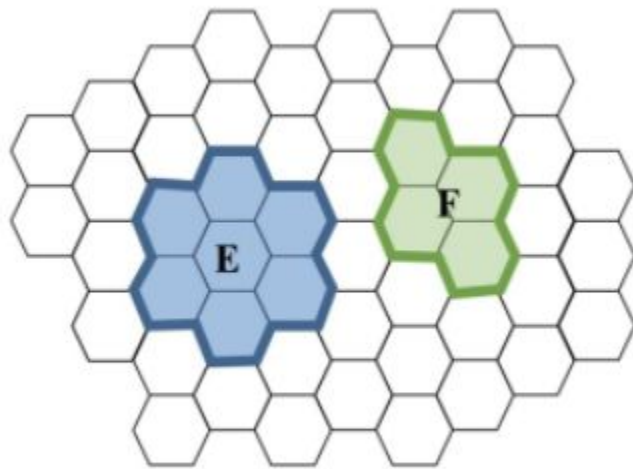
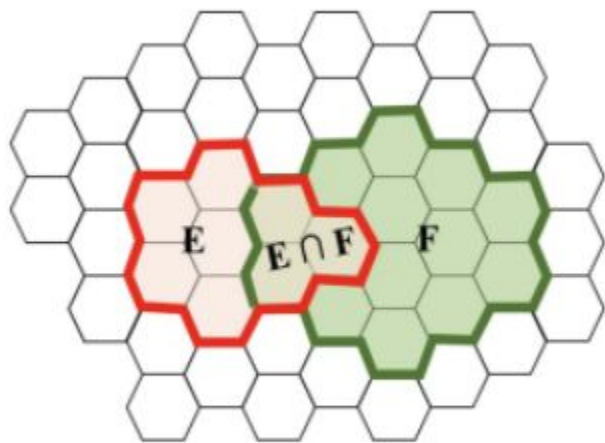
Solving Bayes Problems: A Recipe

1. Define events with letters (Z = has Zika, T = tests positive)
2. Write the goal: $P(Z \mid T)$
3. Inventory what you know: prior $P(Z)$, likelihoods $P(T \mid Z)$, $P(T \mid Z^c)$
4. Apply Bayes: $P(Z \mid T) = P(T \mid Z) P(Z) / P(T)$
5. Get $P(T)$ from LOTP

Someone ask me a
question

Independence

Mutually Exclusive or Independent?



Independent vs. Mutually Exclusive

Independent: $P(E \cap F) = P(E) \cdot P(F)$, or equivalently $P(E | F) = P(E)$

Learning F tells you nothing about E

Mutually exclusive: $E \cap F = \emptyset$, so $P(E \cap F) = 0$

Learning F tells you E did NOT happen

Implies $P(E \cup F) = P(E) + P(F)$

If $P(E) > 0$ and $P(F) > 0$, mutually exclusive events are always dependent

Can we prove this?

Key idea: independent $\rightarrow P(E \cap F) = P(E) P(F)$ mutually exclusive $\rightarrow P(E \cap F) = 0$

A Visual



How to Prove Two Events Independent?

Show that the probability of both together equals the probability of each multiplied!

Assume we have two events A and B — show $P(AB) = P(A) P(B)$

That's it!

Independence: Problem

Roll two dice. E = first die is 1, F = sum is 7, G = sum is 5

Task for you: which are dependent? Which are independent?

Independence: Solution

$$P(E) = 1/6, P(F) = 6/36 = 1/6, P(G) = 4/36 = 1/9$$

$$P(E \cap F) = P\{(1,6)\} = 1/36 = P(E) \cdot P(F) \rightarrow \text{independent}$$

$$P(E \cap G) = P\{(1,4)\} = 1/36 \neq (1/6)(1/9) = 1/54 \rightarrow \text{dependent}$$

Why? Whatever die 1 shows, exactly one value of die 2 makes a sum of 7. For a sum of 5, a 5 or 6 on die 1 makes it impossible.

Inclusion-Exclusion

$$P(E_1 \text{ or } E_2 \text{ or } \dots \text{ or } E_n) = \sum_{r=1}^n (-1)^{r+1} Y_r$$

Y_1 = Sum of all events on their own

$$\sum_i P(E_i)$$

Y_2 = Sum of all pairs of events

$$\sum_{i,j} P(E_i \text{ and } E_j) \quad \text{s.t. } i \neq j$$

Y_3 = Sum of all triples of events

$$\sum_{i,j,k} P(E_i \text{ and } E_j \text{ and } E_k) \quad \text{s.t. } i \neq j, j \neq k, i \neq k$$

Where Y_r is the sum, for all combinations of r events, of the probability of the intersection of those events

Review Problems

Frisbee Problem

You flip **two frisbees**. For each frisbee, the probability that it lands “**heads**” is **0.6**. The two frisbees are considered “**even**” if both frisbees are heads **or** both frisbees are tails.

What is the probability that the frisbees are even?

Frisbee Problem: Solution

Let H_1, H_2 = frisbee 1, 2 lands heads. The two flips are independent.

Even = (both heads) OR (both tails). These can't both happen $\rightarrow$ mutually exclusive $\rightarrow$ just add (easy OR)

$$P(\text{both heads}) = P(H_1) \cdot P(H_2) = 0.6 \cdot 0.6 = 0.36 \quad (\text{easy AND})$$

$$P(\text{both tails}) = P(H_1^c) \cdot P(H_2^c) = 0.4 \cdot 0.4 = 0.16$$

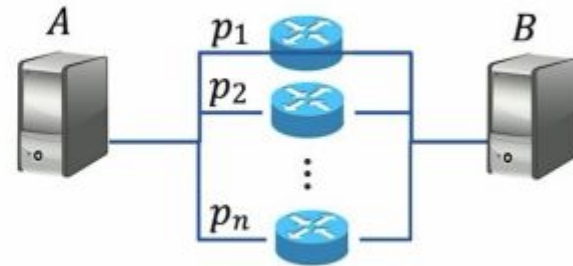
$$\text{Answer: } P(\text{even}) = 0.36 + 0.16 = 0.52$$

Network Reliability

Consider the following parallel network:

- n independent routers, each with probability p_i of functioning (where $1 \leq i \leq n$)
- E = functional path from A to B exists.

What is $P(E)$?



Network Reliability: Solution

E = at least one router works: an OR of n overlapping events (hard OR)

Flip it (De Morgan): E^c = every router fails: an AND of independent events (easy AND)

$$P(E^c) = (1 - p_1)(1 - p_2) \cdots = \prod_i (1 - p_i) \quad (\text{product over } i = 1, \dots, n)$$

Sanity check: 3 routers, each $p_i = 0.9 \rightarrow P(E) = 1 - (0.1)^3 = 0.999$

$$\text{Answer: } P(E) = 1 - \prod_i (1 - p_i)$$

Zika Virus...



A patient has a
positive Zika test.

What is the probability they have zika?

-
- *0.8% of people have zika*
 - *Test has 90% positive rate for people with zika*
 - *Test has 7% positive rate for people without zika*

Zika Virus: Solution

Define events: Z = has Zika, T = tests positive. Want $P(Z \mid T)$.

Know: $P(Z) = 0.008$ (prior), $P(T \mid Z) = 0.90$, $P(T \mid Z^c) = 0.07$

Step 1 – LOTP for the denominator:

$$\begin{aligned} P(T) &= P(T \mid Z) \cdot P(Z) + P(T \mid Z^c) \cdot P(Z^c) \\ &= (0.90)(0.008) + (0.07)(0.992) \\ &= 0.0072 + 0.06944 = 0.07664 \end{aligned}$$

$$P(\text{zika} \mid \text{positive}) = P(\text{positive} \mid \text{zika}) P(\text{zika}) / P(\text{positive})$$

Answer: $P(\text{Zika} \mid \text{positive test}) \approx 0.094 \approx 9\%$

More Practice

Warm-up 1: 8-Sided Die

Roll a fair 8-sided die (faces 1 through 8).

A = the result is even. B = the result is ≥ 6 .

(a) What is $P(A)$?

(b) What is $P(B)$?

(c) What is $P(A \cap B)$?

(d) What is $P(A \cup B)$?

Hint: write out the outcomes in each event.

Warm-up 1: Solution

$S = \{1, \dots, 8\}$, all equally likely, so $P(E) = |E| / 8$

(a) $A = \{2, 4, 6, 8\} \rightarrow P(A) = 4/8 = 1/2$

(b) $B = \{6, 7, 8\} \rightarrow P(B) = 3/8$

(c) $A \cap B = \{6, 8\} \rightarrow P(A \cap B) = 2/8 = 1/4$

(d) Inclusion–exclusion: $P(A \cup B) = 1/2 + 3/8 - 1/4 = 5/8$

Check by listing: $A \cup B = \{2, 4, 6, 7, 8\}$, 5 outcomes $\rightarrow 5/8$ ✓

Answers: (a) $1/2$ (b) $3/8$ (c) $1/4$ (d) $5/8$

Warm-up 2: Birthday Party

You have 6 friends: George, Hannah, Kevin, Shree, Ulo, and Sam.

(a) How many ways are there to choose 2 of the 6 friends to invite?

(b) Shree and Sam must come together or not at all. Now how many ways are there to choose 2 friends?

Warm-up 2: Solution

(a) Order doesn't matter: $C(6, 2) = 6! / (2! \cdot 4!) = 15$

(b) Split into cases that can't overlap (sum rule):

Case 1: invite the pair {Shree, Sam} $\rightarrow$ 1 way

Case 2: invite neither $\rightarrow$ choose 2 of the other 4: $C(4, 2) = 6$

Total = $1 + 6 = 7$

(Inviting only one of Shree and Sam isn't allowed.)

Answers: (a) 15 (b) 7

Warm-up 3: Entry Codes

Entry codes are 6 digits and may start with 0 (e.g., 379009, 012345, 000000).

- (a) How many codes are possible?
- (b) How many are the same two digits repeated three times (121212, 333333)?
- (c) How many use each of the digits 1–6 exactly once (254631)?
- (d) Stretch: a code is valid if its digits sum to exactly 9. What is $P(\text{valid})$ for a code picked uniformly at random?

Warm-up 3: Solution (1/2)

- (a) 10 choices for each of 6 positions (product rule): $10^6 = 1,000,000$
- (b) The first two digits fix the whole code: $10 \cdot 10 = 100$
- (c) Arrange 6 distinct digits in a row: $6! = 720$
- (Careful in (b): 333333 counts once, as the pair “33”.)

Warm-up 3: Solution (2/2)

(d) Count digit strings with $d_1 + d_2 + \dots + d_6 = 9$, where each digit ≥ 0

No digit can exceed 9, since the total is only 9

Stars and bars: 9 stars and 5 bars in a row $\rightarrow C(14, 5) = 2002$

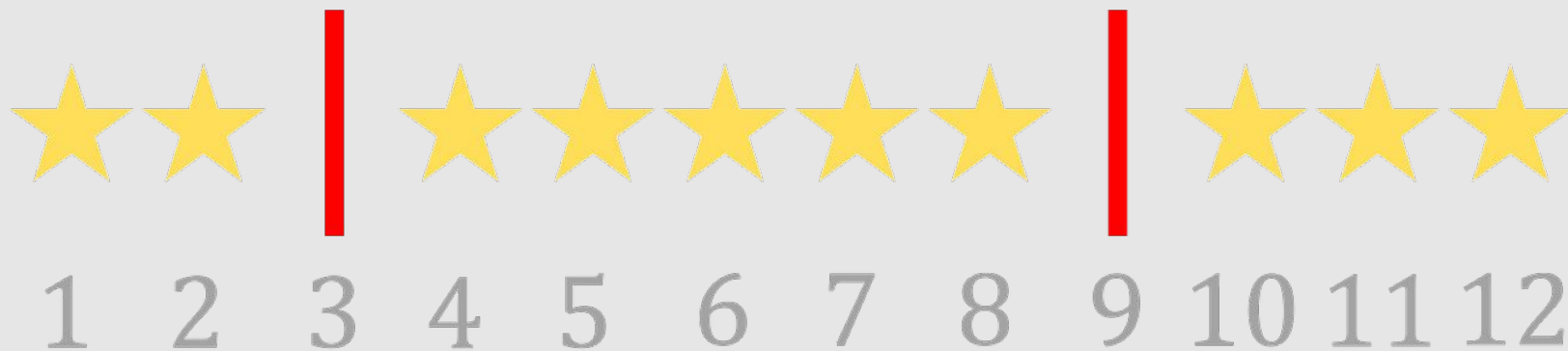
$$P(\text{valid}) = 2002 / 10^6 \approx 0.002$$

(Checked by brute force over all 10^6 codes.)

Answers: (a) 10^6 (b) 100 (c) 720 (d) $2002 / 10^6 \approx 0.002$

Extra (for this week): Stars and Bars

10 identical units among 3 labeled bins



$(10 + (3-1))$ choose 2

Counting Challenge: Six-Card Poker

A casino deals 6-card poker hands from a standard 52-card deck (hands are unordered).

(a) Three pair: 2 cards of one rank, 2 of another, and 2 of a third. What is $P(\text{three pair})$?

(b) Pyramid: 3 cards of one rank, 2 of another, and 1 of a third. What is $P(\text{pyramid})$?

Solution: Six-Card Poker

Sample Space: $|S| = C(52, 6) = 20,358,520$ unordered hands

(a) Three pair: Choose 3 ranks, then choose 2 suits for each rank:

$$|E| = C(13, 3) \cdot C(4, 2)^3 = 286 \cdot 216 = 61,776 \rightarrow P(\text{three pair}) = 61,776 / 20,358,520 \approx \mathbf{0.3034\%}$$

(b) Pyramid (3–2–1): Choose distinct ranks for triple, pair, single ($13 \cdot 12 \cdot 11$), then pick suits:

$$|E| = (13)(12)(11) \cdot C(4, 3) \cdot C(4, 2) \cdot C(4, 1) = 1,716 \cdot 96 = 164,736 \rightarrow P(\text{pyramid}) = 164,736 / 20,358,520 \approx \mathbf{0.8092\%}$$

Answer: $P(\text{three pair}) \approx 0.3034\%$, $P(\text{pyramid}) \approx 0.8092\%$

Warm-up 4: Poker Straight

Draw a 5-card hand from a standard 52-card deck.

A straight is 5 cards of consecutive ranks, e.g., 2-3-4-5-6. An ace can be low (A-2-3-4-5) or high (10-J-Q-K-A). Suits don't matter (count straight flushes too).

What is $P(\text{straight})$?

First, describe the sample space and the event in words.

Warm-up 4: Solution

S = all 5-card hands (unordered): $|S| = C(52, 5) = 2,598,960$

E = all straights. Build one step by step (product rule):

Pick the lowest rank (A, 2, ..., 10): 10 ways, which fixes all 5 ranks

Pick a suit for each card: $4^5 = 1024$ ways

$$|E| = 10 \cdot 1024 = 10,240 \rightarrow P(\text{straight}) = 10,240 / 2,598,960 \approx 0.0039$$

Why not $\times 2$ for “up or down”? Naming a straight by its lowest card counts each hand exactly once.

Answer: $P(\text{straight}) = 10,240 / 2,598,960 \approx 0.0039$

Warm-up 5: The Casino Game

Place 2 tokens on dice sums (2–12). Both can go on the same number. The dealer rolls two dice twice. On each roll, if a token is on that sum, one token is removed and you win \$1000.

- (a) Strategy (7, 7): find $P(\text{win } \$2000)$, $P(\text{win } \$0)$, and $P(\text{win } \$1000)$.
- (b) Strategy (2, 7): same questions.
- (c) Preview of expectation: which strategy wins more on average?

Warm-up 5: Solution (1/2)

(a) Strategy (7, 7): $P(\text{sum} = 7) = 6/36 = 1/6$ on each roll, and the rolls are independent

$$P(\$2000) = (1/6)(1/6) = 1/36$$

$$P(\$0) = (5/6)(5/6) = 25/36$$

$$P(\$1000) = 1 - 1/36 - 25/36 = 10/36 \quad (\text{complement!})$$

$$\text{Average winnings: } 2000 \cdot (1/36) + 1000 \cdot (10/36) \approx \$333.33$$

Warm-up 5: Solution (2/2)

(b) Strategy (2, 7): $P(2) = 1/36$, $P(7) = 6/36$

\$2000: one roll is a 2 and the other a 7: $2 \cdot (1/36)(6/36) = 1/108$

\$0: neither roll is a 2 or a 7: $(29/36)^2 = 841/1296$

\$1000: $1 - 1/108 - 841/1296 = 443/1296$

(c) Average: $2000 \cdot (1/108) + 1000 \cdot (443/1296) \approx \$360.34 > \$333.33$

Two different numbers cover more outcomes (best: 6 & 7, $\approx \$564$)

Answer (c): strategy (2, 7) wins more on average: $\approx \$360$ vs. $\approx \$333$

Conditional 1: Three Coin Flips

You flip a fair coin 3 times.

Given that you saw at least one head, what is the probability that you saw at least two heads?

Conditional 1: Solution

S has $2^3 = 8$ equally likely outcomes

F = at least one head = everything except TTT $\rightarrow |F| = 7$

E = at least two heads = {HHT, HTH, THH, HHH} $\rightarrow |E| = 4$, all inside F

$$P(E | F) = |E \cap F| / |F| = 4/7 \approx 0.571$$

Compare: $P(E) = 4/8 = 1/2$, so learning F raised the probability

Answer: $P(\text{at least 2 heads} \mid \text{at least 1 head}) = 4/7 \approx 0.571$

Conditional 2: Two Dice

Roll two fair dice. X = the first die, Y = the second die.

- (a) What is $P(X > 3 \mid Y = 2)$?
- (b) What is $P(X = 4 \mid X + Y = 8)$?

Conditional 2: Solution

(a) The dice are independent, so knowing Y tells us nothing about X :

$$P(X > 3 \mid Y = 2) = P(X > 3) = 3/6 = 1/2$$

(b) Shrink S to $X + Y = 8$: $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$

5 equally likely outcomes, and only $(4,4)$ has $X = 4 \rightarrow 1/5$

By definition: $P(X = 4, Y = 4) / P(X + Y = 8) = (1/36)/(5/36) = 1/5$

Answers: (a) 1/2 (b) 1/5

Conditional 3: 8-Sided Die, Continued

Same die: $A = \text{even}$, $B = \geq 6$. From Warm-up 1: $P(A) = 1/2$, $P(B) = 3/8$, $P(A \cap B) = 1/4$

- (a) What is $P(A \mid B)$? (b) What is $P(B \mid A)$?
- (c) Is $P(A \cap B) = P(A) P(B)$? Always, never, or sometimes?
- (d) Is $P(A \cap B) = P(A \mid B) P(B)$? Always, never, or sometimes?
- (e) $C = \text{the result is a cube (1 or 8)}$. What is $P(C \mid A \cap B)$?

Conditional 3: Solution

- (a) $P(A \mid B) = P(A \cap B) / P(B) = (1/4) / (3/8) = 2/3$ (in $\{6, 7, 8\}$, the evens are 6 and 8)
- (b) $P(B \mid A) = (1/4) / (1/2) = 1/2$, which is not $P(A \mid B)$!
- (c) Not here: $P(A) P(B) = 3/16 \neq 1/4$. Sometimes: true only if A and B are independent
- (d) Always true: this is the chain rule (the definition, rearranged)
- (e) $A \cap B = \{6, 8\}$, and only 8 is a cube $\rightarrow P(C \mid A \cap B) = 1/2$

Answers: (a) 2/3 (b) 1/2 (c) sometimes (d) always (e) 1/2

Chain Rule 1: The Late Cat (Part 1)

It rains (R) $1/3$ of the time. Heavy traffic (T) happens with probability $1/2$ if it rains, $1/4$ if not.

A cat drives to work. It is late (L) with probability $1/2$ if there is rain and traffic, $1/8$ if there is neither, and $1/4$ otherwise.

(a) On a random day, what is the probability it is not raining, there is heavy traffic, and the cat isn't late?

Chain Rule 1: Solution

Want $P(R^c T L^c)$. Chain rule: each factor conditions on the ones before it

$$P(R^c T L^c) = P(R^c) \cdot P(T \mid R^c) \cdot P(L^c \mid R^c T)$$

$$P(R^c) = 1 - 1/3 = 2/3$$

$$P(T \mid R^c) = 1/4$$

$$P(L^c \mid R^c T) = 1 - 1/4 = 3/4 \quad (\text{the “otherwise” case, complemented})$$

$$\text{Answer: } P(R^c T L^c) = (2/3)(1/4)(3/4) = 1/8$$

Chain Rule 2: White Elephant

n people each bring a gift. One at a time, each person is randomly assigned a gift from the pile (possibly their own).

- (a) What is the probability that all n people get their own gift?
- (b) Exactly $n - 1$ people?
- (c) Exactly $n - 2$ people?
- (d) You may take your turn whenever you like. Which turn maximizes your chance of getting your own gift?

Chain Rule 2: Solution (1/2)

(a) Chain rule: $P(\text{person 1 gets theirs}) = 1/n$, then $1/(n - 1)$, ...

$$P = (1/n) \cdot (1/(n - 1)) \cdot \dots \cdot (1/1) = 1/n!$$

(b) 0: a wrong gift belongs to someone else, who then can't get theirs

(c) Exactly one pair swaps; everyone else is forced

$$C(n, 2) \text{ ways out of } n! \rightarrow P = C(n, 2) / n! = 1 / (2 (n - 2)!)$$

Chain Rule 2: Solution (2/2)

(d) It doesn't matter: your chance is always $1/n$

Go first: $1/n$

Wait k turns: your gift must survive them, then you must draw it

$$P = (n - k)/n \cdot 1/(n - k) = 1/n \quad (\text{chain rule again!})$$

Intuition: your gift is equally likely to be anywhere in the order

Bonus: $P(\text{nobody gets theirs}) = 1/3$ ($n = 3$), $3/8$ ($n = 4$), $\rightarrow 1/e$

Answers: (a) $1/n!$ (b) 0 (c) $1 / (2 (n - 2)!)$ (d) any turn: always $1/n$

Independence 1: Penny and Nickel

Flip one penny and one nickel.

A: the penny comes up tails

B: the nickel comes up tails

C: neither coin comes up tails

D: exactly one coin comes up tails

Which pairs of these events, if any, are independent?

Independence 1: Solution

$S = \{HH, HT, TH, TT\}$ (penny first). $P(A) = P(B) = P(D) = 1/2$, $P(C) = 1/4$

A, B: $P(AB) = P(TT) = 1/4 = (1/2)(1/2) \rightarrow$ independent

A, D: $P(AD) = P(TH) = 1/4 = (1/2)(1/2) \rightarrow$ independent (surprise!)

B, D: $P(BD) = P(HT) = 1/4 = (1/2)(1/2) \rightarrow$ independent

A & C, B & C, C & D: disjoint, so $P = 0 \neq$ product $\rightarrow$ dependent

Takeaway: check the definition, because intuition can mislead

Answer: A & B, A & D, B & D are independent; every pair with C is not

Independence 2: mRNA Patterns

An mRNA string has 8 letters. Each letter is generated independently:
 $P(A) = P(U) = 1/3$, $P(C) = P(G) = 1/6$.

V_i = the pattern AGUA starts at position i .

- (a) What is $P(V_1)$?
- (b) Are V_1 and V_2 independent?
- (c) Are V_1 and V_4 independent?
- (d) Are V_1 and V_5 independent?

Independence 2: Solution

(a) The letters are independent: $P(V_1) = (1/3)(1/6)(1/3)(1/3) = 1/162$

(b) Dependent: V_1 puts a G in position 2, so AGUA can't start there

$$P(V_2 \mid V_1) = 0 \neq 1/162$$

(c) Dependent: V_1 puts an A in position 4, giving V_4 a head start

$$P(V_4 \mid V_1) = P(G, U, A \text{ in } 5-7) = (1/6)(1/3)(1/3) = 1/54 > 1/162$$

(d) Independent: positions 1–4 and 5–8 don't overlap, and letters are independent

Answers: (a) 1/162 (b) dependent (c) dependent (d) independent

Independence 3 (Challenge): Three Events

For events A, B, and C:

A and C are independent; B and C are independent

A and B are mutually exclusive

$$P(A \cup C) = 2/3, \quad P(B \cup C) = 3/4, \quad P(A \cup B \cup C) = 11/12$$

Find $P(A)$, $P(B)$, and $P(C)$.

Independence 3: Solution (1/2)

Let $a = P(A)$, $b = P(B)$, $c = P(C)$

Independence: $P(AC) = ac$, $P(BC) = bc$. Disjoint: $P(AB) = P(ABC) = 0$

Inclusion–exclusion gives three equations:

$$(1) \quad a + c - ac = 2/3$$

$$(2) \quad b + c - bc = 3/4$$

$$(3) \quad a + b + c - ac - bc = 11/12$$

Independence 3: Solution (2/2)

(1) + (2) - (3): everything cancels except c

$$c = 2/3 + 3/4 - 11/12 = 6/12 = 1/2$$

Plug into (1): $a + 1/2 - a/2 = 2/3 \rightarrow a/2 = 1/6 \rightarrow a = 1/3$

Plug into (2): $b + 1/2 - b/2 = 3/4 \rightarrow b/2 = 1/4 \rightarrow b = 1/2$

Check (3): $1/3 + 1/2 + 1/2 - 1/6 - 1/4 = 11/12 \checkmark$

Answer: $P(A) = 1/3$, $P(B) = 1/2$, $P(C) = 1/2$

LOTP 1: Two-Step Marbles

Step 1: draw a marble from a bag of 1 red, 2 green, and 1 blue marble. Its color picks a bucket.

Step 2: draw a marble from that bucket. $P(\text{blue marble})$ is:

1/4 in the red bucket, 0 in the green bucket, 1/3 in the blue bucket

A = the step-2 marble is blue.

(a) What is $P(A)$?

(b) Given A , what is the probability we used the red bucket?

LOTP 1: Solution (1/2)

Partition on the step-1 color: $P(R) = 1/4$, $P(G) = 2/4 = 1/2$, $P(B) = 1/4$

Likelihoods: $P(A | R) = 1/4$, $P(A | G) = 0$, $P(A | B) = 1/3$

(a) LOTP: $P(A) = P(A | R) P(R) + P(A | G) P(G) + P(A | B) P(B)$

$$= (1/4)(1/4) + (0)(1/2) + (1/3)(1/4)$$

$$= 1/16 + 0 + 1/12 = 7/48 \approx 0.146$$

LOTP 1: Solution (2/2)

$$\begin{aligned} \text{(b) Bayes: } P(R \mid A) &= P(A \mid R) P(R) / P(A) \\ &= (1/16) / (7/48) = 3/7 \approx 0.43 \end{aligned}$$

Prior $P(R) = 1/4 \rightarrow$ posterior $3/7$: seeing blue makes “red bucket” more likely

Check: $P(B \mid A) = (1/12) / (7/48) = 4/7$ and $P(G \mid A) = 0$

$$3/7 + 4/7 + 0 = 1 \quad \checkmark$$

Answers: (a) $P(A) = 7/48 \approx 0.146$ (b) $P(R \mid A) = 3/7 \approx 0.43$

LOTP 2: The Late Cat (Part 2)

Same setup as Chain Rule 1:

$$P(R) = 1/3; P(T \mid R) = 1/2, P(T \mid R^c) = 1/4$$

$$P(L \mid R T) = 1/2, P(L \mid R^c T^c) = 1/8, \text{ and } 1/4 \text{ otherwise}$$

(b) What is the probability the cat is late?

(c) The manager knows the cat was late today. What is the probability it rained?

LOTP 2: Solution (1/2)

(b) Partition on the 4 (rain, traffic) cases. Chain rule on each branch:

$$R\ T: \quad (1/3)(1/2) = 1/6 \rightarrow \text{late: } (1/6)(1/2) = 1/12$$

$$R\ T^c: \quad (1/3)(1/2) = 1/6 \rightarrow \text{late: } (1/6)(1/4) = 1/24$$

$$R^c\ T: \quad (2/3)(1/4) = 1/6 \rightarrow \text{late: } (1/6)(1/4) = 1/24$$

$$R^c\ T^c: \quad (2/3)(3/4) = 1/2 \rightarrow \text{late: } (1/2)(1/8) = 1/16$$

$$\text{LOTP: } P(L) = 1/12 + 1/24 + 1/24 + 1/16 = 11/48 \approx 0.229$$

LOTP 2: Solution (2/2)

(c) Bayes: $P(R \mid L) = P(R \cap L) / P(L)$

$P(R \cap L)$ = the two rainy branches = $1/12 + 1/24 = 1/8$

$P(R \mid L) = (1/8) / (11/48) = 6/11 \approx 0.545$

Prior $1/3 \rightarrow$ posterior $6/11$: a late cat is evidence of rain

Sanity check: the answer is between 0 and 1, and it went up, as expected

Answers: (b) $P(L) = 11/48 \approx 0.229$ (c) $P(R \mid L) = 6/11 \approx 0.545$

Bayes 1: SARS Test

A test detects SARS 98% of the time when a person has it. It has a 1% false-positive rate. 0.5% of the population has SARS.

- (a) You test positive. What is $P(\text{SARS} \mid \text{positive})$?
- (b) You test negative. What is $P(\text{SARS} \mid \text{negative})$?

Bayes 1: Solution (1/2)

F = has SARS, E = tests positive. Prior: $P(F) = 0.005$

Likelihoods: $P(E \mid F) = 0.98$, $P(E \mid F^c) = 0.01$

LOTP: $P(E) = (0.98)(0.005) + (0.01)(0.995) = 0.0049 + 0.00995 = 0.01485$

(a) $P(F \mid E) = 0.0049 / 0.01485 \approx 0.330$

With 1000 people: about 5 have SARS and test positive, and about 10 healthy people test positive $\rightarrow$ about $5 / 15 \approx 1/3$

Bayes 1: Solution (2/2)

(b) Now the evidence is E^c : $P(E^c | F) = 0.02$, $P(E^c | F^c) = 0.99$

$$P(E^c) = (0.02)(0.005) + (0.99)(0.995) = 0.0001 + 0.98505 = 0.98515$$

$$P(F | E^c) = 0.0001 / 0.98515 \approx 0.0001$$

A positive test raises the risk about 66× (0.5% → 33%)

A negative test nearly rules SARS out

Base rates matter!

Answers: (a) $P(\text{SARS} | +) \approx 0.33$ (b) $P(\text{SARS} | -) \approx 0.0001$

Bayes 2: Algorithmic Fairness

A protected demographic D makes up 10% of the population. Within D , 5% have a health condition (H). Outside D , only 1% do.

(a) What is $P(H \mid D)$?

(b) What is $P(D \mid H)$?

(c) A naive model always predicts “no condition.” How often is it correct on a representative sample?

Bayes 2: Solution (1/2)

(a) Given directly: $P(H \mid D) = 0.05$

(b) Bayes: $P(D \mid H) = P(H \mid D) P(D) / P(H)$

$$\text{LOTP: } P(H) = (0.05)(0.10) + (0.01)(0.90) = 0.005 + 0.009 = 0.014$$

$$P(D \mid H) = 0.005 / 0.014 = 5/14 \approx 0.357$$

Check with 10,000 people: 50 in D have H; 90 outside D have H

$$P(D \mid H) = 50 / (50 + 90) = 5/14 \checkmark$$

Bayes 2: Solution (2/2)

(c) The model is right exactly when $H = 0$:

$$P(H = 0) = 1 - 0.014 = 0.986 \rightarrow 98.6\% \text{ accurate}$$

...but it never helps a single person with the condition!

Takeaway: accuracy alone can hide serious problems

Note: $P(H \mid D) = 0.05$ but $P(D \mid H) \approx 0.36$, so $P(A \mid B) \neq P(B \mid A)$

Answers: (a) 0.05 (b) $5/14 \approx 0.357$ (c) 98.6% accurate

For Next Time

FILL OUT FORM

Will post notes!

Can also make summary notes from class if useful...

PSET 1 due Oct 5

Midterm review: during class? Or additional time?